

Forward and Backward Passes of a Log-Barrier ADMM QP Solver

A perturbed-KKT derivation in the TurboMPC ADMM convention

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1 From trajectory optimization to a QP

We consider the discrete-time optimal control problem (OCP) in the form of [1],

$$\begin{aligned} \min_{(x_t, u_t)_{t=0}^N} \quad & \sum_{t=0}^{N-1} \ell_t(x_t, u_t) + \ell_N(x_N) \\ \text{s.t.} \quad & x_0 = x_{\text{init}}, \\ & f_t(x_t, u_t, x_{t+1}, u_{t+1}) = 0, \quad t = 0, \dots, N-1, \\ & \underline{g}_t \leq g_t(x_t, u_t) \leq \bar{g}_t, \quad t = 0, \dots, N, \end{aligned} \quad (1)$$

with states $x_t \in \mathbb{R}^{n_x}$, controls $u_t \in \mathbb{R}^{n_u}$, implicit discretized dynamics f_t , and pointwise-in-time inequality constraints g_t (bounds, obstacles, friction cones, ...). The OCP data depend on parameters θ (cost weights, constraint geometry) with respect to which we ultimately want gradients.

Stacked NLP. Overloading $x := (x_t, u_t)_{t=0}^N \in \mathbb{R}^{n_w}$ for the stacked stage variables, collect the cost into ℓ , the initial-condition and dynamics residuals into $f \in \mathbb{R}^m$, and the inequalities into *one-sided* rows $g \in \mathbb{R}^p$ (each finite bound of (1) contributes one row, $g_t \leq \bar{g}_t$ or $-g_t \leq -\underline{g}_t$; infinite bounds are dropped), so that (1) is the parametric nonlinear program

$$\min_x \ell(x; \theta) \quad \text{s.t.} \quad f(x; \theta) = 0, \quad g(x; \theta) \leq h(\theta). \quad (2)$$

SQP. Sequential quadratic programming [8, Ch. 18] solves (2) by iterating: at the current primal–dual iterate, quadraticize the Lagrangian, linearize the constraints, solve the resulting QP, and update the iterate along its solution with a linesearch. With the constraint Jacobians $C := \partial f / \partial x$ and $G := \partial g / \partial x$ evaluated at the iterate, $P \approx \nabla_{xx}^2 \mathcal{L}$ a Hessian approximation (in practice Gauss–Newton), and q the corresponding gradient, each subproblem is the convex QP studied in the rest of this note:

$$\text{QP}(\theta) : \quad \min_x \frac{1}{2} x^\top P x + q^\top x \quad \text{s.t.} \quad Cx = c, \quad Gx \leq h, \quad (3)$$

with $P \succeq 0$, $C \in \mathbb{R}^{m \times n_w}$, $G \in \mathbb{R}^{p \times n_w}$. The time structure of (1) makes P block-tridiagonal, C block-bidiagonal, and G block-diagonal [1] — structure both passes below

exploit. A differentiable MPC layer must provide (i) a *forward pass* producing the primal–dual solution of (3) at each SQP iteration (Section 3) and (ii) a *backward pass* producing $\nabla_\theta L$ for a scalar loss L of the converged solution — obtained by differentiating the *NLP*’s optimality system, not the QP’s (Section 4).

2 Perturbed KKT and smoothed complementarity

With multipliers $y_f \in \mathbb{R}^m$ (equalities) and $y_g \in \mathbb{R}^p$ (inequalities) and the slack $s := h - Gx$, the KKT conditions of (3) are

$$\begin{aligned} Px + q + C^\top y_f + G^\top y_g &= 0, & Cx &= c, \\ Gx + s &= h, & s &\geq 0, \quad y_g \geq 0, \quad s \circ y_g = 0, \end{aligned} \quad (4)$$

where $\circ$ is the elementwise product. All difficulty — combinatorial for the solver, and geometric for differentiation — sits in the last line. The solution map $\theta \mapsto (x, y_f, y_g)$ of (4) is only piecewise smooth: at points where *strict complementarity* fails ($s_i = y_{g,i} = 0$ for some row) the KKT Jacobian is singular and the map is nondifferentiable, with gradient jumps across active-set changes [6, 7]. We therefore smooth (4) with two perturbations, used together or separately.

(i) Barrier perturbation κ . Following the interior-point tradition, replace complementarity by the *perturbed KKT* (central-path) condition [8, Ch. 19]

$$s \circ y_g = \kappa \mathbf{1}, \quad s > 0, \quad y_g > 0, \quad (5)$$

for a barrier parameter $\kappa > 0$. This is the stationarity system of the equality-constrained log-barrier subproblem (add $-\kappa \sum_i \log s_i$ to the cost); as $\kappa \searrow 0$ it recovers (4). Differentiating the κ -perturbed system instead of (4) is exactly the mechanism used to obtain smooth gradients through QP/NLP layers [5, 7].

(ii) Inequality-slack (Moreau) perturbation (δ_ξ, γ_ξ) . Following TurboMPC’s slack convention [1], introduce an *inequality slack* $\xi \geq 0$ per one-sided row, toggled by the binary indicator $\delta_\xi \in \{0, 1\}$ and priced quadratically: $Gx - \delta_\xi \xi \leq h$ with penalty $\delta_\xi \frac{\gamma_\xi}{2} \|\xi\|^2$,

$\gamma_\xi > 0$. For $\delta_\xi = 1$, minimizing over ξ first replaces the indicator of $\{Gx \leq h\}$ by its Moreau envelope $\frac{\gamma_\xi}{2} \text{dist}^2(Gx, \{\cdot \leq h\})$ [4]: the constraint wall becomes a stiff quadratic ramp, the QP stays feasible under SQP linearization error, and the multiplier ramps up continuously as $y_g = \gamma_\xi \xi$ (TurboMPC’s `ADMMSlack` smoothing [1]). **The default is** $\delta_\xi = 0$ (no inequality slack); ($\delta_\xi=1, \gamma_\xi \rightarrow \infty$) is equivalent to $\delta_\xi = 0$.

Combined perturbed system. Applying both, the log-barrier subproblem is

$$\begin{aligned} \min_{x, \xi} \quad & \frac{1}{2} x^\top P x + q^\top x + \delta_\xi \frac{\gamma_\xi}{2} \|\xi\|^2 - \kappa \sum_{i=1}^p \log s_i \\ \text{s.t.} \quad & Cx = c, \quad s = h - Gx + \delta_\xi \xi, \end{aligned} \quad (6)$$

whose stationarity in s gives $y_g = \kappa/s$ (elementwise) and, when $\delta_\xi = 1$, stationarity in ξ gives $\gamma_\xi \xi = y_g$. Eliminating $\xi = \gamma_\xi^{-1} y_g$, the optimality conditions collapse to a square smooth root-finding problem in $w := (x, y_f, y_g)$:

$$F_{\text{QP}}(w; \theta) := \begin{bmatrix} Px + q + C^\top y_f + G^\top y_g \\ Cx - c \\ s \circ y_g - \kappa \mathbf{1} \end{bmatrix} = 0, \quad (7)$$

$$s := h - Gx + \delta_\xi \gamma_\xi^{-1} y_g (> 0), \quad y_g > 0. \quad (8)$$

The forward pass converges to the root of (7) for each QP subproblem; the backward pass differentiates the nonlinear counterpart of this system (Section 4). The two perturbations place the solver in a four-way family of ADMM methods:

	$\kappa = 0$ (hard compl.)	$\kappa > 0$ (central path)
$\delta_\xi = 0$	KKT (4); OSQP [2]	pure barrier (default)
$\delta_\xi = 1$	Moreau; <code>ADMMSlack</code> [1]	slack barrier

Both $\kappa > 0$ cells are modes of this solver (Section 3). For any $\kappa > 0$ the residual F_{QP} is smooth and its Jacobian is nonsingular (Section 4), so $\theta \mapsto w(\theta)$ is C^1 — including across activation/deactivation of constraints. The price is an $O(\kappa)$ bias of $x(\theta)$ relative to the hard QP solution, and an $O(\|y_g\|/\gamma_\xi)$ constraint violation from the inequality slack when $\delta_\xi = 1$; $\kappa \rightarrow 0$ (with $\delta_\xi = 0$, or $\gamma_\xi \rightarrow \infty$) recovers (4).

3 Forward pass: ADMM with a log-barrier slack update

We follow the ADMM convention of [1] (itself OSQP’s splitting [2, 3]). Stack the constraints as $A := \begin{bmatrix} C \\ G \end{bmatrix}$ and give Ax its own copy $z = (z_f, z_g)$:

$$\begin{aligned} \min_{x, z, \xi} \quad & \frac{1}{2} x^\top P x + q^\top x + \delta_\xi \frac{\gamma_\xi}{2} \|\xi\|^2 + \mathcal{I}_{\{c\}}(z_f) + B_\kappa(z_g, \xi) \\ \text{s.t.} \quad & Ax - z = 0, \end{aligned} \quad (9)$$

where $\mathcal{I}$ denotes an indicator function and

$$B_\kappa(z_g, \xi) := -\kappa \sum_{i=1}^p \log(h_i - z_{g,i} + \delta_\xi \xi_i) \quad (10)$$

is the log barrier on the copy. (At $\kappa = 0$, B_κ degenerates to the indicator of $\{z_g - \delta_\xi \xi \leq h\}$ and (9) is exactly the TurboMPC splitting.) Introducing duplicated variables $(\tilde{x}, \tilde{z})$ as in [1], multipliers w for $\tilde{x} - x = 0$ (which vanishes identically after the first iteration and is dropped below) and $y = (y_f, y_g)$ for $\tilde{z} - z = 0$, and step-size parameters $\sigma > 0$, $\rho := \text{blkdiag}(\rho_f I_m, \rho_g I_p) \succ 0$, the augmented Lagrangian is

$$\begin{aligned} \mathcal{L}_{\sigma, \rho} = \quad & \frac{1}{2} \tilde{x}^\top P \tilde{x} + q^\top \tilde{x} + \delta_\xi \frac{\gamma_\xi}{2} \|\xi\|^2 + \mathcal{I}_{\{c\}}(z_f) + B_\kappa(z_g, \xi) \\ & + \mathcal{I}_{\{0\}}(A\tilde{x} - \tilde{z}) + \frac{\sigma}{2} \|\tilde{x} - x\|^2 + \frac{1}{2} \|\tilde{z} - z + \rho^{-1} y\|_\rho^2, \end{aligned} \quad (11)$$

with $\|v\|_\rho^2 := v^\top \rho v$. One ADMM iteration minimizes $\mathcal{L}_{\sigma, \rho}$ over $(\tilde{x}, \tilde{z})$, then over (x, z, ξ) , then ascends the duals, with over-relaxation $\alpha \in (0, 2)$.

3.1 Primal update

Minimizing (11) over $(\tilde{x}, \tilde{z})$ is an equality-constrained strictly convex QP whose KKT conditions are the linear system [1, Eq. (7)]

$$\begin{bmatrix} P + \sigma I & A^\top \\ A & -\rho^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}^{k+1} \\ \nu^{k+1} \end{bmatrix} = \begin{bmatrix} \sigma x^k - q \\ z^k - \rho^{-1} y^k \end{bmatrix}. \quad (12)$$

The $-\rho^{-1}$ block is eliminated by a Schur complement:

$$S \tilde{x}^{k+1} = \sigma x^k - q + A^\top (\rho z^k - y^k), \quad \tilde{z}^{k+1} = A \tilde{x}^{k+1}, \quad (13)$$

with $S := P + \sigma I + A^\top \rho A \succ 0$. By the structure noted in Section 1, S is block-tridiagonal in time; its sparse factorization is computed once and reused across iterations, refactorizing only when ρ is adapted [1, 2]. The over-relaxed iterates are

$$\hat{x}^{k+1} = \alpha \tilde{x}^{k+1} + (1 - \alpha) x^k, \quad \hat{z}^{k+1} = \alpha \tilde{z}^{k+1} + (1 - \alpha) z^k, \quad (14)$$

and the x -minimization gives $x^{k+1} = \hat{x}^{k+1}$.

3.2 Slack update: the proximal operator of the log barrier

Minimizing (11) over z_f gives $z_f^{k+1} = c$ (so z_f need not be stored). The inequality block is the essential step. Let

$$\hat{z}_g^{k+1} := \hat{z}_g^{k+1} + \rho_g^{-1} y_g^k \quad (15)$$

be the usual ADMM target — the same point OSQP projects onto $\{z_g \leq h\}$ and TurboMPC’s `ADMMSlack` projects smoothly. Here the update is the prox of the barrier, jointly over (z_g, ξ) and separable across rows:

$$(z_g, \xi)^{k+1} = \arg \min_{z, \xi} B_\kappa(z, \xi) + \delta_\xi \frac{\gamma_\xi}{2} \|\xi\|^2 + \frac{\rho_g}{2} \|z - \hat{z}_g^{k+1}\|^2 \quad (16)$$

(for $\delta_\xi = 0$ the inequality slack is fixed, $\xi \equiv 0$, exactly as in TurboMPC’s slack update). Per row, with the barrier slack $s(z, \xi) := h - z + \delta_\xi \xi$, the objective of (16) is

$$\varphi(z, \xi) = -\kappa \log s(z, \xi) + \delta_\xi \frac{\gamma_\xi}{2} \xi^2 + \frac{\rho_g}{2} (z - \tilde{z}_g^{k+1})^2. \quad (17)$$

This is an *unconstrained* smooth problem: the barrier absorbs $s \geq 0$ into its domain ($\varphi \rightarrow +\infty$ as $s \rightarrow 0^+$), and the quadratic terms make φ strictly convex and coercive there, so the unique minimizer is the zero-gradient point. With $\partial s / \partial z = -1$ and $\partial s / \partial \xi = \delta_\xi$, the chain rule gives

$$\begin{aligned} \partial_z \varphi &= \frac{\kappa}{s} + \rho_g (z - \tilde{z}_g^{k+1}) = 0 \quad \Rightarrow \quad \frac{\kappa}{s} = \rho_g (\tilde{z}_g^{k+1} - z), \\ \partial_\xi \varphi &= -\frac{\kappa}{s} + \gamma_\xi \xi = 0 \quad (\delta_\xi=1) \quad \Rightarrow \quad \gamma_\xi \xi = \frac{\kappa}{s}, \end{aligned} \quad (18)$$

and the stationary $\xi = \kappa / (\gamma_\xi s) > 0$ is automatically positive, so $\xi \geq 0$ never activates. Solving (18) for $z = \tilde{z}_g^{k+1} - \kappa / (\rho_g s)$ and $\delta_\xi \xi = \delta_\xi \kappa / (\gamma_\xi s)$ and substituting back into $s = h - z + \delta_\xi \xi$ with the residual $r^{k+1} := h - \tilde{z}_g^{k+1}$ yields $s = r^{k+1} + \Gamma / s$, i.e. the scalar quadratic $s^2 - r^{k+1} s - \Gamma = 0$ per row, whose positive root is, elementwise,

$$\begin{cases} s^{k+1} = b_\Gamma(r^{k+1}), & \Gamma := \kappa(\rho_g^{-1} + \delta_\xi \gamma_\xi^{-1}), \\ \xi^{k+1} = \delta_\xi \frac{\kappa}{\gamma_\xi s^{k+1}}, & z_g^{k+1} = h - s^{k+1} + \xi^{k+1}, \end{cases} \quad (19)$$

where b_μ is the *retraction map* (positive root of $b^2 - vb - \mu = 0$)

$$b_\mu(v) := \frac{v + \sqrt{v^2 + 4\mu}}{2}, \quad \mu > 0, \quad (20)$$

with the elementary properties:

- (i) $b_\mu(v) > 0$ for all $v \in \mathbb{R}$;
- (ii) $b_\mu(v) b_\mu(-v) = \mu$ (built-in complementarity);
- (iii) $b_\mu(v) - b_\mu(-v) = v$;
- (iv) $b_\mu(v) \rightarrow \max(v, 0)$ as $\mu \rightarrow 0^+$ (smooth ReLU);
- (v) $b'_\mu(v) = \frac{1}{2}(1 + v/\sqrt{v^2 + 4\mu}) \in (0, 1)$ and $b'_\mu(v) + b'_\mu(-v) = 1$.

In the default mode $\delta_\xi = 0$: $\Gamma = \kappa / \rho_g$, $\xi^{k+1} = 0$, and $z_g^{k+1} = h - b_{\kappa/\rho_g}(r^{k+1})$. (For $v < 0$ we evaluate $b_\mu(v) = 2\mu / (\sqrt{v^2 + 4\mu} - v)$ to avoid numerical issues.)

3.3 Dual update and exact perturbed complementarity

The dual ascent uses the over-relaxed copy [1]:

$$y_f^{k+1} = y_f^k + \rho_f (\hat{z}_f^{k+1} - c), \quad y_g^{k+1} = y_g^k + \rho_g (\hat{z}_g^{k+1} - z_g^{k+1}). \quad (21)$$

By (15), $y_g^{k+1} = \rho_g (\tilde{z}_g^{k+1} - z_g^{k+1})$, and the z -stationarity in (18) evaluates this in closed form:

$$y_g^{k+1} = \frac{\kappa}{s^{k+1}} \Rightarrow s^{k+1} \circ y_g^{k+1} = \kappa \mathbf{1}, \quad \gamma_\xi \xi^{k+1} = \delta_\xi y_g^{k+1} \quad (22)$$

elementwise, *at every iteration* k — not only in the limit. The iterates live on the central-path manifold (5) by construction, with $y_g^{k+1} > 0$ automatic from property (i), and the inequality-slack stationarity holds identically in both modes. At a fixed point, $Ax = z$ makes $s = h - Gx + \delta_\xi \xi = h - Gx + \delta_\xi \gamma_\xi^{-1} y_g$, so (x, y_f, y_g) satisfies *exactly* the perturbed KKT system (7)–(8) of the QP.

Limits. As $\kappa \rightarrow 0^+$, property (iv) collapses (19), for $\delta_\xi = 1$, to TurboMPC’s smoothed projection $\tilde{\Pi}^{\gamma_\xi}(\tilde{z}_g) = \frac{\rho_g}{\gamma_\xi + \rho_g} \tilde{z}_g + \frac{\gamma_\xi}{\gamma_\xi + \rho_g} \Pi(\tilde{z}_g)$ [1, Eq. (15)], and, for $\delta_\xi = 0$, to OSQP’s hard projection $z_g^{k+1} = \min(\tilde{z}_g^{k+1}, h)$: a smooth deformation of the standard projection ADMM, at identical per-iteration cost.

Termination, ρ -adaptation, continuation. Convergence is declared on the standard residuals $r_{\text{prim}} = \|Ax - z\|_\infty$ and $r_{\text{dual}} = \|Px + q + A^\top y\|_\infty$ [1, Eq. (3)] (slack stationarity $\gamma_\xi \xi - \delta_\xi y_g = 0$ holds identically by (22)); $\bar{\rho}$ is adapted OSQP-style from the residual ratio with $\rho_f = 10^3 \bar{\rho}$, $\rho_g = \bar{\rho}$, $\sigma = 10^{-6}$, refactorizing S on each change. Optionally κ is annealed geometrically with warm starts; for differentiable use a fixed κ is preferred so the backward differentiates the fixed point the forward reached.

4 Backward pass: implicit differentiation of the NLP’s perturbed KKT

Let $L = L(x^*)$ be a scalar loss of the converged SQP solution. We do *not* differentiate through solver iterations — neither the ADMM loop nor the SQP loop. Nor do we differentiate the last QP subproblem’s optimality system: the object that carries the sensitivities of (2) is the *NLP’s* own smoothed KKT system, and QP-level sensitivities coincide with the NLP’s only when the QP carries the *exact* Lagrangian Hessian at the solution [7, Thm. 2]. Since the forward SQP typically runs a Gauss–Newton P (Section 1), differentiating the QP it actually solved can silently degrade the gradients [7, Rem. 3]. The backward pass therefore re-assembles the NLP system at the converged point.

4.1 The NLP’s smoothed KKT system

At SQP convergence, the iterate $w^* := (x^*, y_f, y_g)$ satisfies the $(\kappa, \delta_\xi, \gamma_\xi)$ -perturbed optimality conditions of (2) — the nonlinear counterpart of (7), with the smoothed complementarity imposed on the *nonlinear* residual (the interior-point-smoothed conditions of [7, Eq. (10)], whose

barrier parameter τ is our κ):

$$F(w; \theta) := \begin{bmatrix} \nabla_x \ell(x; \theta) + C(x)^\top y_f + G(x)^\top y_g \\ f(x; \theta) \\ s \circ y_g - \kappa \mathbf{1} \end{bmatrix} = 0, \quad (23)$$

$$s := h(\theta) - g(x; \theta) + \delta_\xi \gamma_\xi^{-1} y_g (> 0), \quad y_g > 0, \quad (24)$$

where $C(x) := \partial f / \partial x$ and $G(x) := \partial g / \partial x$ are the constraint Jacobians evaluated at x (for affine f, g and quadratic ℓ , (23) reduces exactly to (7)). The first row is ∇_x of the NLP Lagrangian $\mathcal{L} := \ell(x; \theta) + y_f^\top f(x; \theta) + y_g^\top g(x; \theta)$. Differentiating (23) at its root yields the *exact* gradient of the relaxed solution map, smooth across active-set changes.

4.2 The implicit function theorem

The residual $F(w; \theta)$ of (23) is C^1 in (w, θ) whenever ℓ, f, g are twice continuously differentiable [7, Ass. 1]. Let w^* satisfy $F(w^*; \theta) = 0$ with $s > 0$, $y_g > 0$ (guaranteed for $\kappa > 0$ by (22); at SQP convergence the linearized and nonlinear residuals coincide), and define $M := \partial F / \partial w|_{(w^*, \theta)}$. If M is nonsingular (established below), the implicit function theorem yields a locally unique C^1 map $\theta \mapsto w(\theta)$ with $F(w(\theta); \theta) \equiv 0$ and, by total differentiation $\frac{\partial F}{\partial w} \frac{\partial w}{\partial \theta} + \frac{\partial F}{\partial \theta} = 0$,

$$\frac{\partial w}{\partial \theta} = -M^{-1} \frac{\partial F}{\partial \theta} \quad (25)$$

— the sensitivity result of [7, Thm. 1]; their Thm. 3 adds that the smoothed map is C^1 for small κ and approaches the unsmoothed one with error $O(\kappa)$. Rather than forming $\partial w / \partial \theta$ column-by-column (n_θ solves with M), the gradient of a scalar loss needs a *single* solve with $M^\top$: absorbing the sign, define the adjoint $\lambda = (\lambda_x, \lambda_f, \lambda_g)$ by

$$M^\top \lambda = -(\nabla_x L, 0, 0), \quad \nabla_\theta L = \left(\frac{\partial F}{\partial \theta} \right)^\top \lambda. \quad (26)$$

4.3 The Jacobian M and the adjoint system

Differentiating (23) with $\partial s / \partial x = -G(x)$ and $\partial s / \partial y_g = \delta_\xi \gamma_\xi^{-1} I$ gives, in block form (rows = residuals, columns = x, y_f, y_g ; all quantities evaluated at w^*),

$$M = \begin{bmatrix} H & C^\top & G^\top \\ C & 0 & 0 \\ -\text{diag}(y_g) G & 0 & \text{diag}(s + \delta_\xi \gamma_\xi^{-1} y_g) \end{bmatrix}, \quad (27)$$

where the (1,1) block is the *exact Lagrangian Hessian* of the NLP,

$$H := \nabla_{xx}^2 \mathcal{L} = \nabla_{xx}^2 \ell + \sum_{j=1}^m y_{f,j} \nabla_{xx}^2 f_j + \sum_{i=1}^p y_{g,i} \nabla_{xx}^2 g_i, \quad (28)$$

not the Gauss–Newton P used by the forward SQP — sensitivities computed with an approximate Hessian can be badly degraded even when the forward solution is accurate [7, Thm. 2, Rem. 3]. Because f and g are stagewise, H retains P ’s block-tridiagonal sparsity. The adjoint system (26) written out is

$$H \lambda_x + C^\top \lambda_f - G^\top \text{diag}(y_g) \lambda_g = -\nabla_x L, \quad (29a)$$

$$C \lambda_x = 0, \quad (29b)$$

$$G \lambda_x + \text{diag}(s + \delta_\xi \gamma_\xi^{-1} y_g) \lambda_g = 0. \quad (29c)$$

Two equivalent reductions of (29) are used in practice.

4.4 Reduction A: retraction parameterization

On the central-path manifold (stated for the default $\delta_\xi = 0$), properties (ii)–(iii) of b_κ make the pair (s, y_g) the graph of a *single* unconstrained variable $v := y_g - s \in \mathbb{R}^p$:

$$s = b_\kappa(-v), \quad y_g = b_\kappa(v), \quad (30)$$

which satisfies $s \circ y_g = \kappa \mathbf{1}$, $s, y_g > 0$ *identically* — the complementarity residual disappears from the system. Differentiating with $B_p := \text{diag}(b'_\kappa(v))$, $B_n := \text{diag}(b'_\kappa(-v))$ (both positive by (v)) gives $dy_g = B_p dv$, $ds = -B_n dv$, and the remaining residuals (stationarity, primal inequality $g(x; \theta) + s - h$, equality) linearize in (dx, dv, dy_f) with matrix $\begin{bmatrix} H & G^\top B_p & C^\top \\ C & -B_n & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Subtracting $G^\top \times$ (second row) from the first and using $B_p + B_n = I$ (property (v)) symmetrizes it:

$$J = \begin{bmatrix} H - G^\top G & G^\top & C^\top \\ G & -B_n & 0 \\ C & 0 & 0 \end{bmatrix} = J^\top. \quad (31)$$

Because J is symmetric, the adjoint solve is

$$J \begin{bmatrix} \lambda_x \\ \lambda_v \\ \lambda_f \end{bmatrix} = - \begin{bmatrix} \nabla_x L \\ 0 \\ 0 \end{bmatrix}, \quad \bar{\lambda}_g := B_p \lambda_v, \quad (32)$$

where $\bar{\lambda}_g$ is the effective inequality adjoint used in the gradient assembly below.

4.5 Reduction B: Schur elimination onto a weighted Hessian

Alternatively — and for either value of δ_ξ — eliminate λ_g directly from (29c): its coefficient $\text{diag}(s + \delta_\xi \gamma_\xi^{-1} y_g)$ is diagonal and strictly positive for $\kappa > 0$, so

$$\lambda_g = -\text{diag}(s + \delta_\xi \gamma_\xi^{-1} y_g)^{-1} G \lambda_x. \quad (33)$$

Substituting into (29a) folds the inequalities into the Hessian block:

$$\begin{bmatrix} H + G^\top \text{diag}(W) G & C^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} \lambda_x \\ \lambda_f \end{bmatrix} = - \begin{bmatrix} \nabla_x L \\ 0 \end{bmatrix}, \quad (34)$$

with the smooth per-row weight

$$W := \frac{y_g}{s + \delta_\xi \gamma_\xi^{-1} y_g}, \quad W|_{\delta_\xi=0} = \frac{y_g}{s} = \frac{y_g^2}{\kappa}, \quad (35)$$

and the effective inequality adjoint $\bar{\lambda}_g := -y_g \circ \lambda_g = \text{diag}(W) G \lambda_x$, matching (32). This reduced saddle system is exactly the symmetric reduced coefficient matrix of [7, Eq. (9)] (there $S^{-1}M = \text{diag}(\mu/s)$ plays the role of W). W is the C^1 analogue of a hard active-set mask:

regime	$y_{g,i}$	s_i	W_i
inactive row	$\rightarrow 0$	> 0	$\rightarrow 0$ (row drops out)
active, $\kappa \rightarrow 0$, $\delta_\xi = 1$	> 0	$\rightarrow \gamma_\xi^{-1} y_{g,i}$	$\rightarrow \gamma_\xi$ (penalty stiffness)
active, $\kappa \rightarrow 0$, $\delta_\xi = 0$	> 0	$\rightarrow 0$	$\rightarrow \infty$ (hard constraint)

so $W \in [0, \gamma_\xi]$ always when $\delta_\xi = 1$. Since G is block-diagonal per stage, $G^\top \text{diag}(W)G$ perturbs only the diagonal blocks of H : (34) has the same block-tridiagonal saddle structure as the forward Schur system (13), and the same structured sparse solver serves both passes [1].

Equivalence of the reductions. Eliminating $\lambda_v = B_n^{-1}G\lambda_x$ from the middle row of (31) gives $H - G^\top G + G^\top B_n^{-1}G = H + G^\top B_p B_n^{-1}G$ (using $I - B_n = B_p$), and on the manifold $B_p B_n^{-1} = \text{diag}(b'_\kappa(v)/b'_\kappa(-v)) = \text{diag}(b_\kappa(v)/b_\kappa(-v)) = \text{diag}(y_g/s)$, which is exactly $W|_{\delta_\xi=0}$; likewise $\bar{\lambda}_g = B_p B_n^{-1}G\lambda_x = \text{diag}(W)G\lambda_x$. (For $\delta_\xi = 1$ the middle block becomes $-(B_n + \gamma_\xi^{-1}B_p)$ and $B_p(B_n + \gamma_\xi^{-1}B_p)^{-1} = \text{diag}(y_g/(s + \gamma_\xi^{-1}y_g)) = W$.) Reductions A and B are the same linear system.

Nonsingularity — why κ is load-bearing. (34) is nonsingular iff C has full row rank (LICQ; true by construction: the initial-condition and dynamics rows) and $H + G^\top \text{diag}(W)G \succ 0$ on $\ker C$, which for small κ follows from the second-order sufficient conditions at the solution [7, Ass. 1, Thm. 1]. The elimination (33) itself required $\text{diag}(s + \delta_\xi \gamma_\xi^{-1} y_g) \succ 0$: for $\kappa > 0$ this holds *everywhere*, because $s \circ y_g = \kappa$ keeps s away from zero. In the hard limit $\kappa = 0$ the genuine failure is a *weakly active* row, $s_i = 0$ and $y_{g,i} = 0$ (strict complementarity fails): the i -th complementarity row of M vanishes identically, M is singular, and the solution map is non-differentiable [6, 7], with $W_i = 0/0$ jumping between 0 (inactive side) and γ_ξ or ∞ (active side). (A strongly active row, $s_i = 0$, $y_{g,i} > 0$, also zeroes the block entry when $\delta_\xi = 0$, but harmlessly: $W_i \rightarrow \infty$ just enforces $G_i \lambda_x = 0$.) The barrier removes the failure: substituting $s = \kappa/y_g$ (elementwise) gives $W = y_g^2/(\kappa + \delta_\xi \gamma_\xi^{-1} y_g^2)$, smooth along the central path, so $\nabla_\theta L$ is continuous across active-set changes [5].

4.6 Assembling the parameter gradients

With (λ_x, λ_f) from (34) (or (32)) and $\bar{\lambda}_g$ from (35), the contraction $\nabla_\theta L = (\partial F / \partial \theta)^\top \lambda$ of (26) takes, at the NLP

level, the compact form

$$\nabla_\theta L = [\partial_\theta \nabla_x \mathcal{L}]^\top \lambda_x + [\partial_\theta f]^\top \lambda_f + [\partial_\theta (g-h)]^\top \bar{\lambda}_g, \quad (36)$$

three mixed vector–Jacobian products of the problem functions evaluated at (w^*, θ) — the adjoint sensitivity of [7, Eq. (13)]. For parameters that enter only the cost (the typical learned-MPC case), $\partial_\theta f = 0$ and $\partial_\theta (g-h) = 0$, and (36) collapses to the single contraction $\nabla_\theta L = [\partial_\theta \nabla_x \ell(x^*; \theta)]^\top \lambda_x$, one vector–Jacobian product of the cost gradient.

QP-layer special case. When the layer is the QP (3) itself, with data $\theta = (P, q, C, c, G, h)$ and $H = P$, expanding (36) datum-by-datum gives the familiar table

$$\begin{aligned} \nabla_P L &= \frac{1}{2}(\lambda_x x^{*\top} + x^* \lambda_x^\top), & \nabla_q L &= \lambda_x, \\ \nabla_C L &= \lambda_f x^{*\top} + y_f \lambda_x^\top, & \nabla_c L &= -\lambda_f, \\ \nabla_G L &= \bar{\lambda}_g x^{*\top} + y_g \lambda_x^\top, & \nabla_h L &= -\bar{\lambda}_g. \end{aligned} \quad (37)$$

Summary. The forward pass drives each QP subproblem to the root of (7) by a projection-free ADMM whose slack update is the closed-form prox of the log barrier (19), holding $s \circ y_g = \kappa \mathbf{1}$ exactly at every iterate. The backward pass differentiates the NLP’s smoothed KKT system (23) at the converged SQP solution with the exact Lagrangian Hessian (28) [7]; the adjoint system (34) shares the block-tridiagonal structure — and the linear solver — of the forward pass. The parameters $(\kappa, \delta_\xi, \gamma_\xi)$ interpolate smoothly between this solver, TurboMPC’s slack-smoothed ADMM, and OSQP’s projection ADMM, keeping the solution map differentiable for every $\kappa > 0$.

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